

# CHOWTA ARAVIND

Software Engineer

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## EDUCATION

**B.Tech in Computer Science & Information Technology CGPA: 9.72 / 10.0**

2022 – 2026

KL University

*Rank #2 in CS & IT Department*

## SKILLS

|                            |                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------|
| <b>Languages &amp; CS</b>  | Java (Primary), JavaScript, Python, C; Data Structures, Algorithms, OOP, Distributed Systems |
| <b>Backend</b>             | Java Spring Boot, Node.js, Java EE, REST API, RAG, Redis, Gradle                             |
| <b>Frontend &amp; DB</b>   | React.js, HTML, CSS, Tailwind, MySQL, MongoDB (NoSQL), VectorDB, DB Indexing                 |
| <b>Tools &amp; Testing</b> | Git, Postman, Agile/Scrum, AWS, Jemmy, K6 (Automated & Performance Testing)                  |

## EXPERIENCE

**Software Developer Intern | Siemens EDA**

July 2025 – Present

- Ranked **Top 2 of 476** in a company-wide innovation program for building an NLP pipeline using RAG, embeddings, and semantic search that cut research time by **70%** and improved retrieval speed **2.5x**.
- Shipped **2+ production features** end-to-end (design through deployment) within Agile sprints, partnering with senior engineers on code review and integration to ship without rework.
- Reduced customer-reported integration issues by **30%** by designing RESTful APIs with versioning, input validation, and security-focused error handling, fixing recurring bugs found during endpoint testing.
- Reduced weekly automated-test failures from **300+ to under 50** by diagnosing intermittent failures across distributed test environments and refactoring framework components to isolate root causes.

## PROJECTS

**AgriConnect Platform** | React, Java Spring Boot, MySQL, Redis

[GitHub](#) ◇ [Article](#)

- Replaced password-based login with **biometric authentication** (Fingerprint API), reducing average sign-in time from **10s to under 2s** on a Spring Boot-based platform connecting farmers with experts.
- Cut API response time from **2.3s to 0.8s** under simulated load by introducing **Redis caching**.
- Improved accessibility for non-English-speaking farmers by building a multilingual engine (i18next) supporting **10+ Indian regional languages**.
- Built a **Decision Tree**-based crop-yield recommendation model trained on the **Kaggle Crop Recommendation dataset (2,200+ records)**, achieving **85% accuracy** on a held-out test set.

**Airline Booking System** | Scalable MERN Application

[GitHub](#)

- Reduced seat fragmentation by **50%** by designing a seat-block allocation algorithm that groups single and double seat requests based on row proximity.
- Validated system reliability under load by performance-testing the booking flow with **K6**, simulating **500+ concurrent users** on booking/search endpoints.
- Cut average search query latency from **3.2s to 1.4s** by adding compound indexes on flight/route fields and tuning MongoDB connection pool size.
- Built an admin dashboard (React + Recharts) surfacing occupancy trends, booking volume, and route-wise demand across **50+ seeded flights** and **1,000+ mock bookings**, with filterable date-range views.

## AWARDS & CERTIFICATIONS

- Top Coder @ KLU** : Top 5 at Codeathon; 500+ DSA problems solved; Rank #2 in CS & IT Department
- Oracle Certified Java SE 11 Developer** [\[Badge\]](#)    **AWS Certified Cloud Practitioner** [\[Badge\]](#)

## LEADERSHIP

**Founder & President, RUBIX Problem-Solving Club**

September 2024 – November 2025

- Built a 50-member technical team delivering coding competitions, workshops, and hackathons; scaled community to **2000+ members** and **1000+ social followers within 6 months**.

**President, EFIT CS&IT Student Body**

November 2025 – March 2026

- Led 2 university-level hackathons (500+ participants each) and 15+ technical events; initiated a placement prep series impacting **70+ attendees per session**.